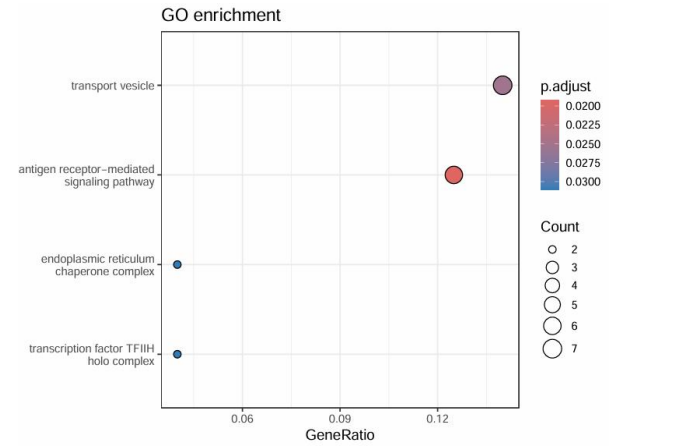
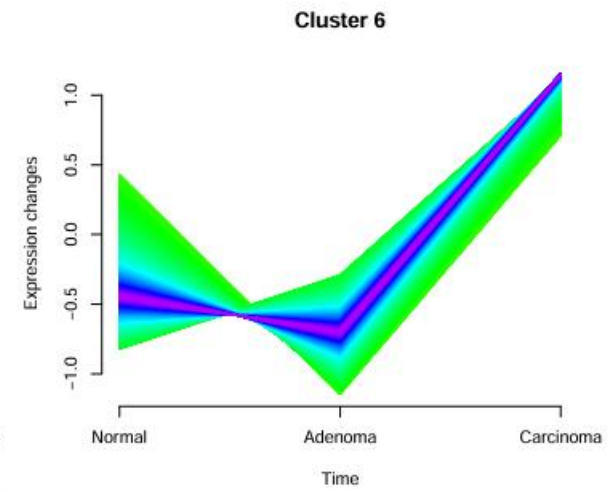
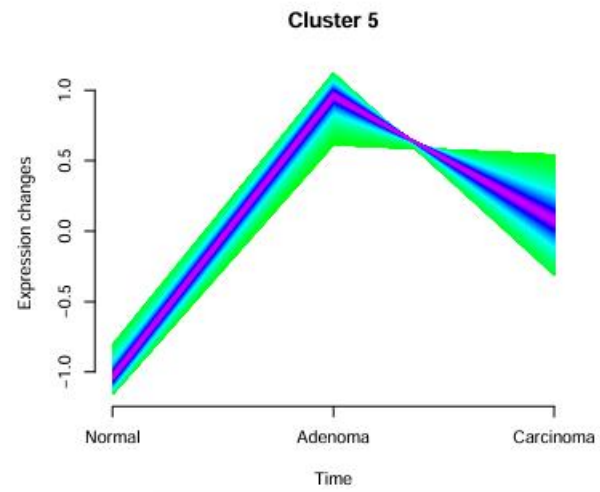
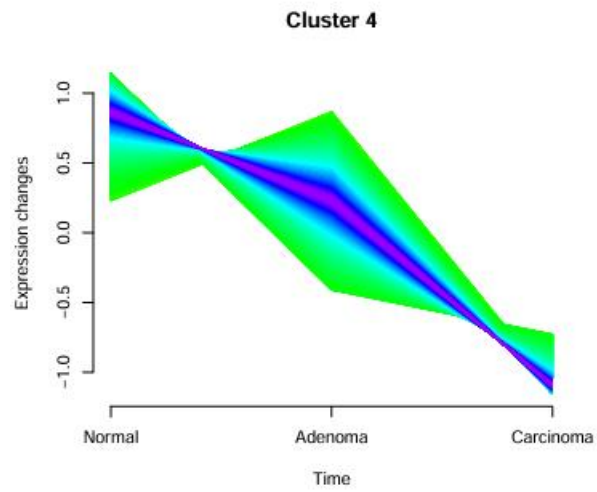
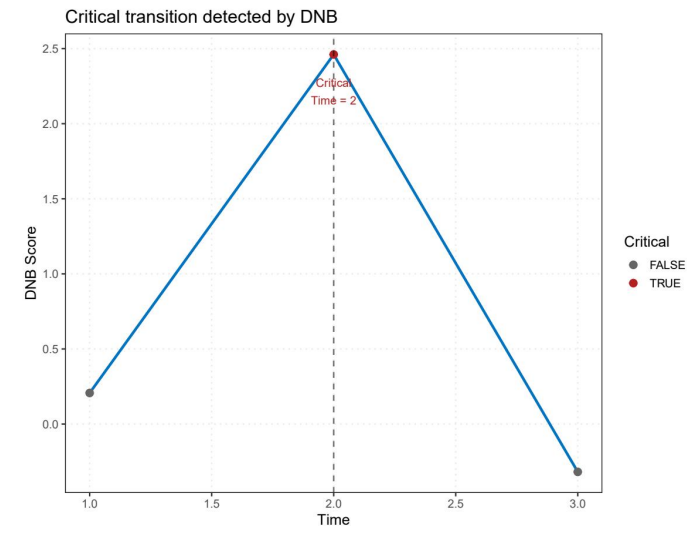
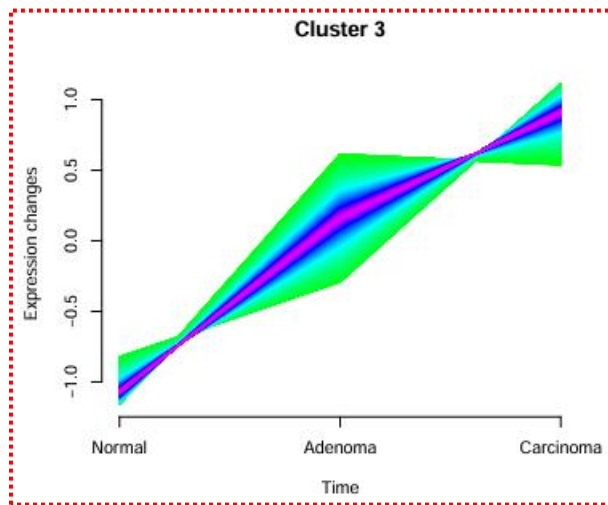
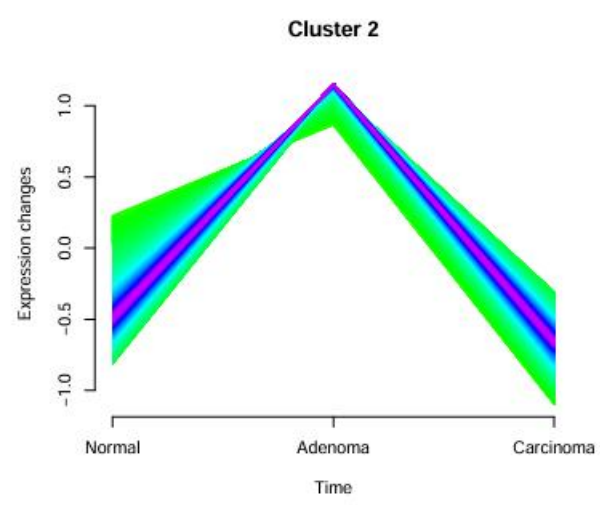
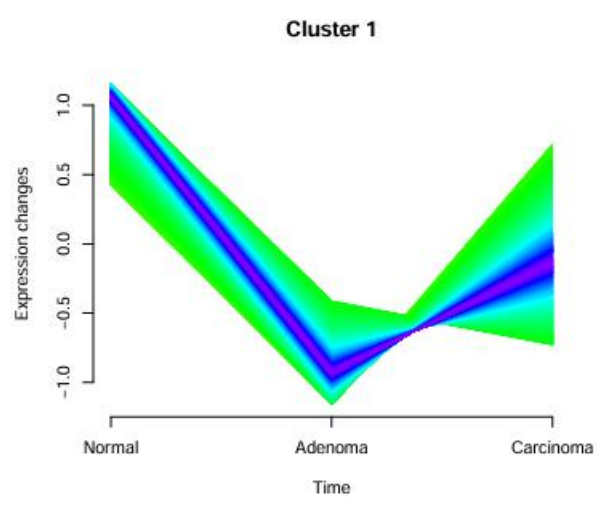
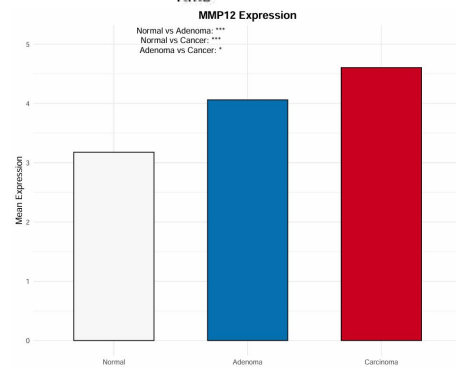


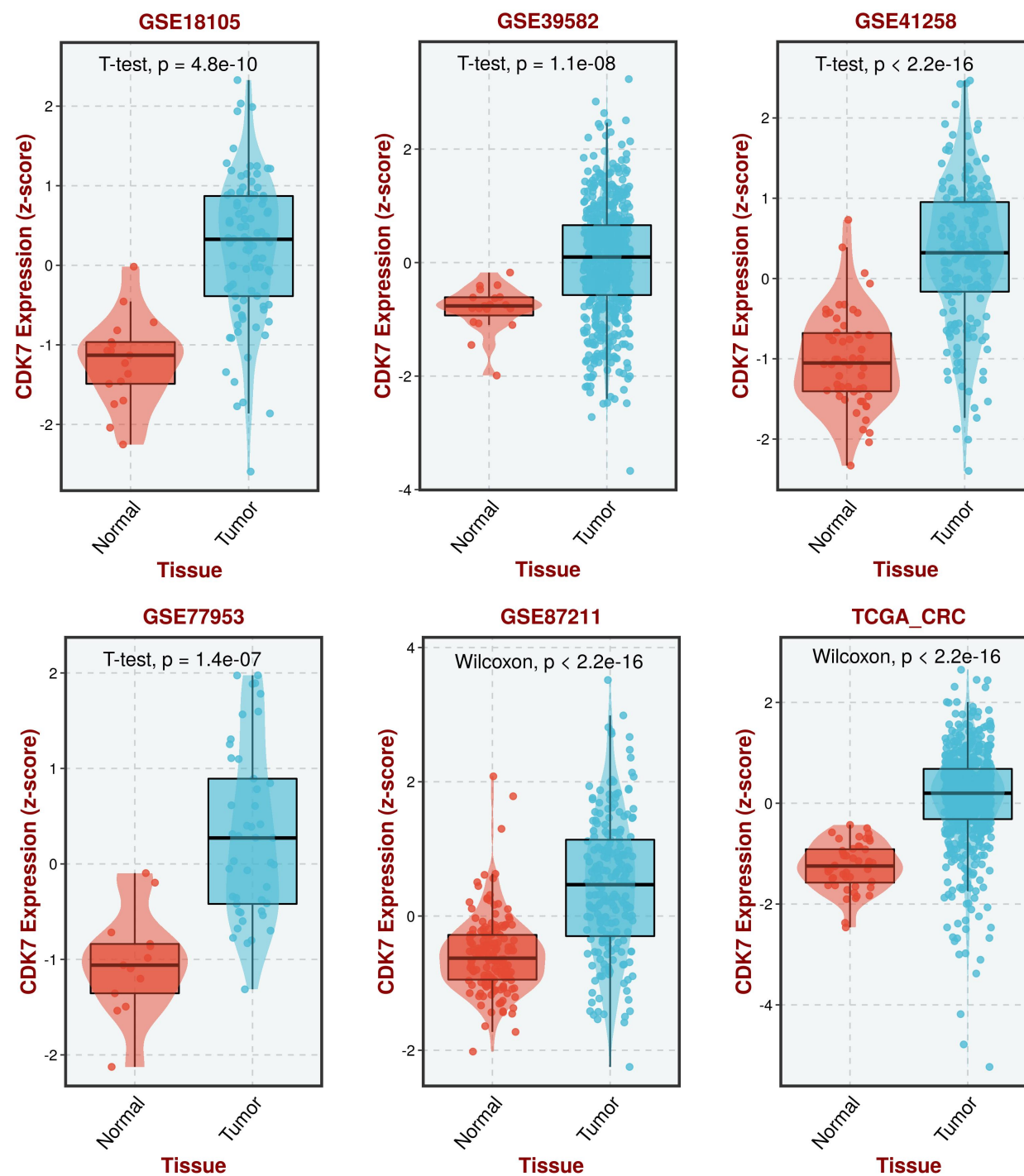


验证集验证取交集



动物模型验证

CDK7



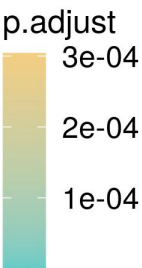
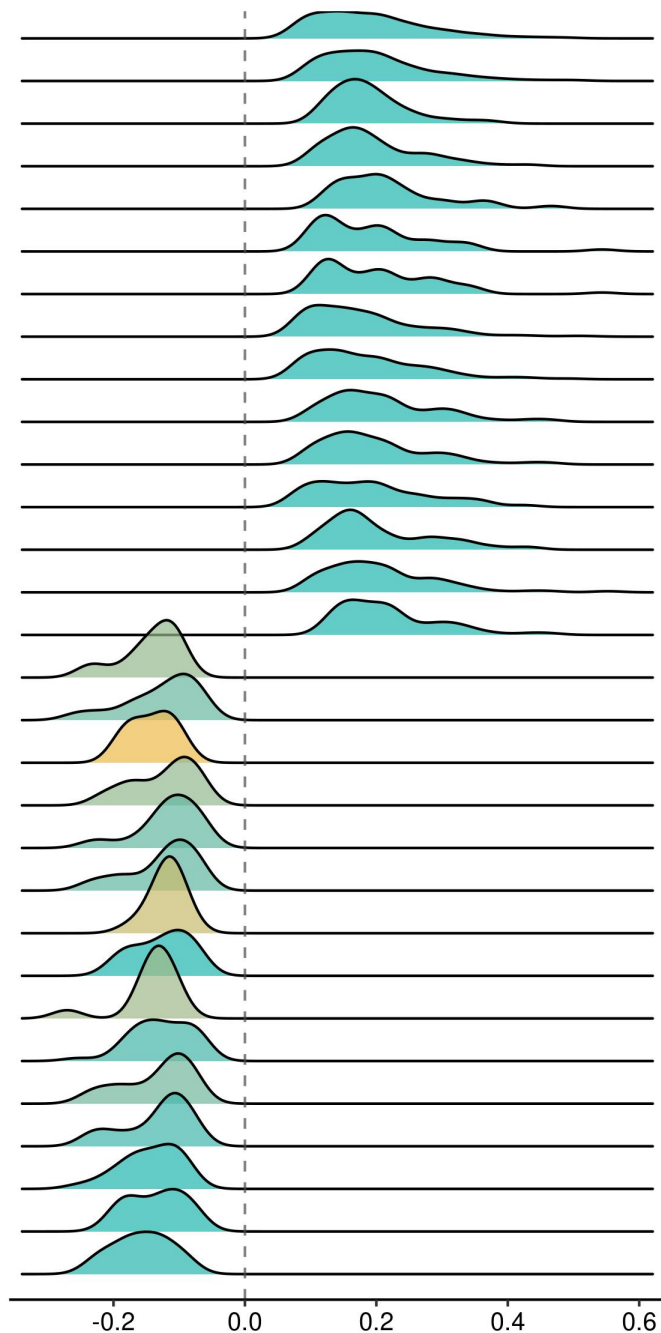
不同数据集中CDK7
在肿瘤中的表达量都
高于正常组织

调控细胞基础代谢与合成功能，参与CRC发生发展

核糖体的生成过程
核糖核蛋白复合物的生成过程
细胞质翻译
线粒体呼吸链复合物的组装
核糖体组装
线粒体翻译
线粒体基因表达
NCRNA 处理
RNA 能量代谢过程
三磷酸腺苷 (ATP) 合成与电子传递耦合
呼吸电子传递链
核糖体大亚基的生成过程
纳德脱氢酶复合物的组装
有氧呼吸
氧化磷酸化

Ribosome biogenesis
Ribonucleoprotein complex biogenesis
Cytoplasmic translation
chondrial respiratory chain complex assembly
Ribosome assembly
Mitochondrial translation
Mitochondrial gene expression
Ncrna processing
Rrna metabolic process
Atp synthesis coupled electron transport
Respiratory electron transport chain
Ribosomal large subunit biogenesis
Nadh dehydrogenase complex assembly
Aerobic respiration
Oxidative phosphorylation
Response to prostaglandin
Collagen fibril organization
Positive regulation of calcium ion import
Negative regulation of interleukin 6 production
Negative regulation of chemotaxis
Semaphorin plexin signaling pathway
Cellular response to zinc ion
Cell cell adhesion via plasma membrane adhesion molecules
Regulation of antigen processing and presentation
Vascular endothelial growth factor receptor signaling pathway
Neuron projection extension involved in neuron projection guidance
Negative chemotaxis
External encapsulating structure organization
Homophilic cell adhesion via plasma membrane adhesion molecules
Apoptotic cell clearance

GSEA-GO Analysis

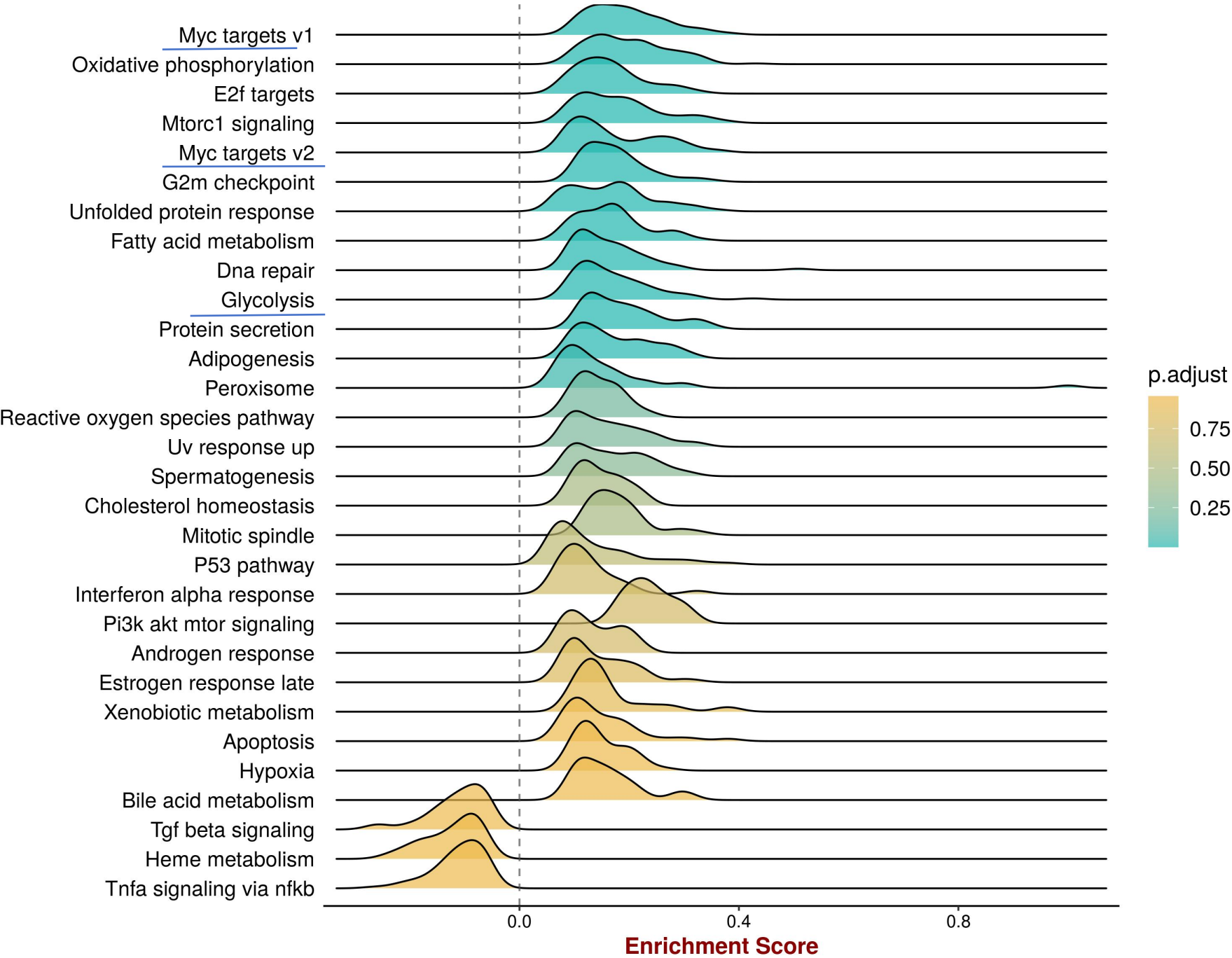


大部分和核糖体线粒体活动呈显著相关性符合恶性细胞发生发展对能量代谢和蛋白质合成增大的需求

参与相关癌症核心特征通路参与癌症进展

疾病的经典恶性机制

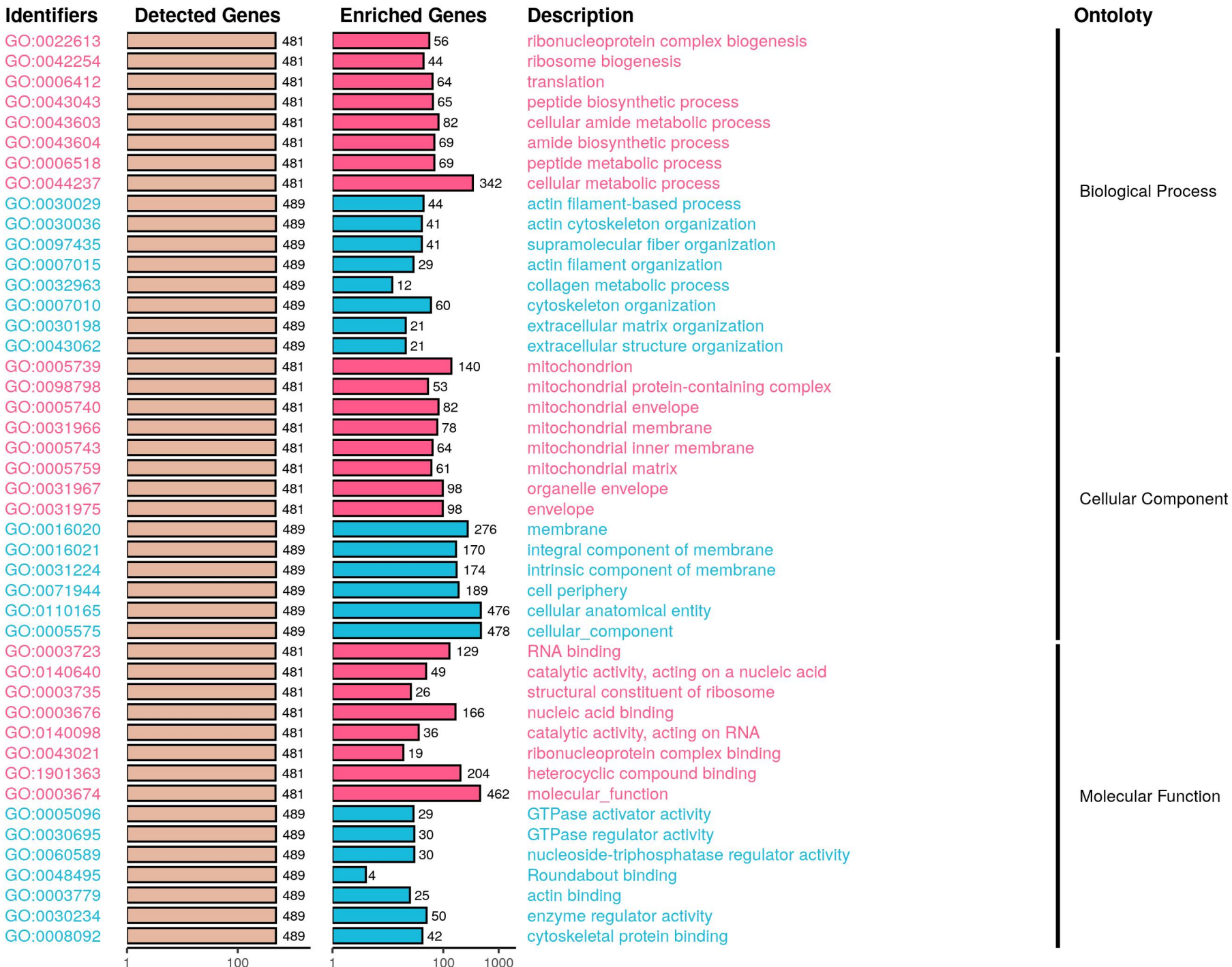
GSEA-Hallmark Analysis



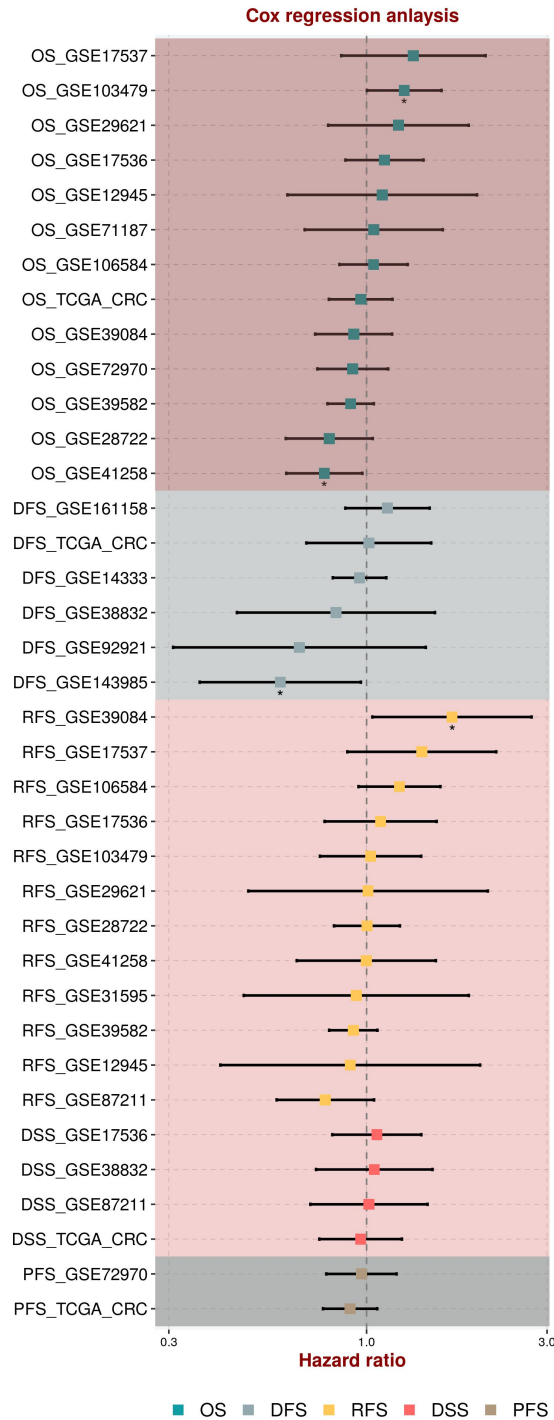
高度富集在 “癌细胞增殖
(Myc/E2f 通路)、癌细胞能量
代谢(糖酵解)、细胞周期调控
(G2m checkpoint)” 等癌症核
心特征通路中

基因功能分类的富集结果表

显著富集于三大类功能，尤其是“线粒体能量代谢、核糖体蛋白合成”等功能

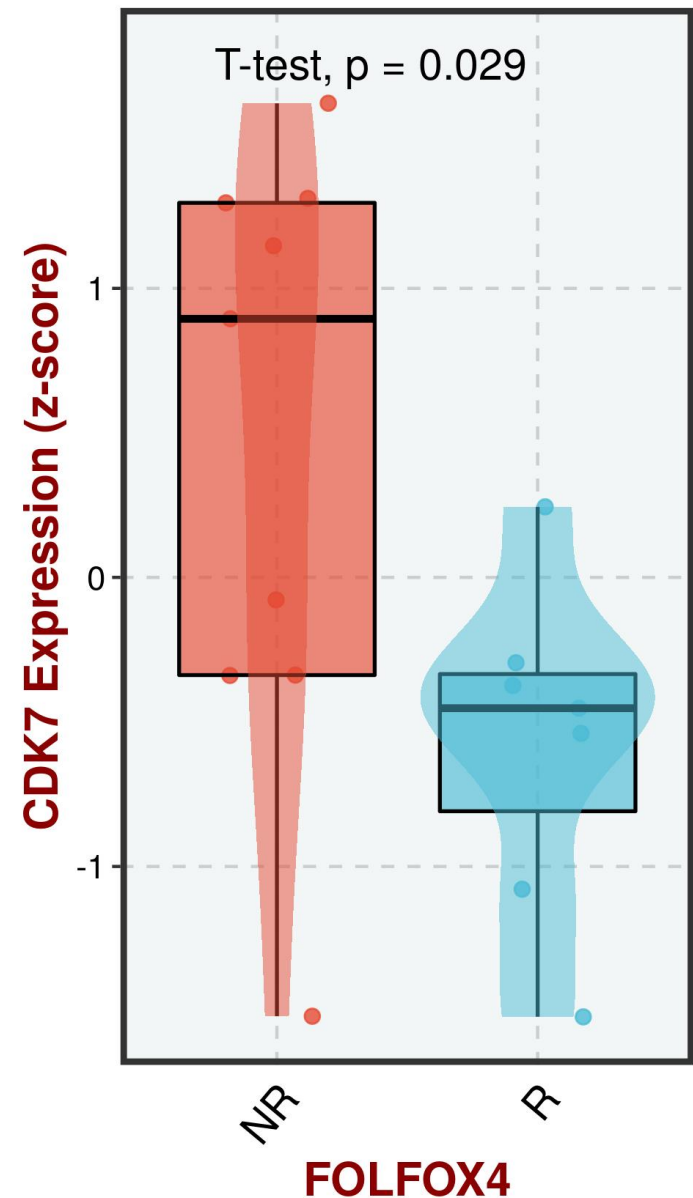


Cox 回归分析图



CDK7 基因在 FOLFOX4 化疗方案下的表达差异

GSE69657

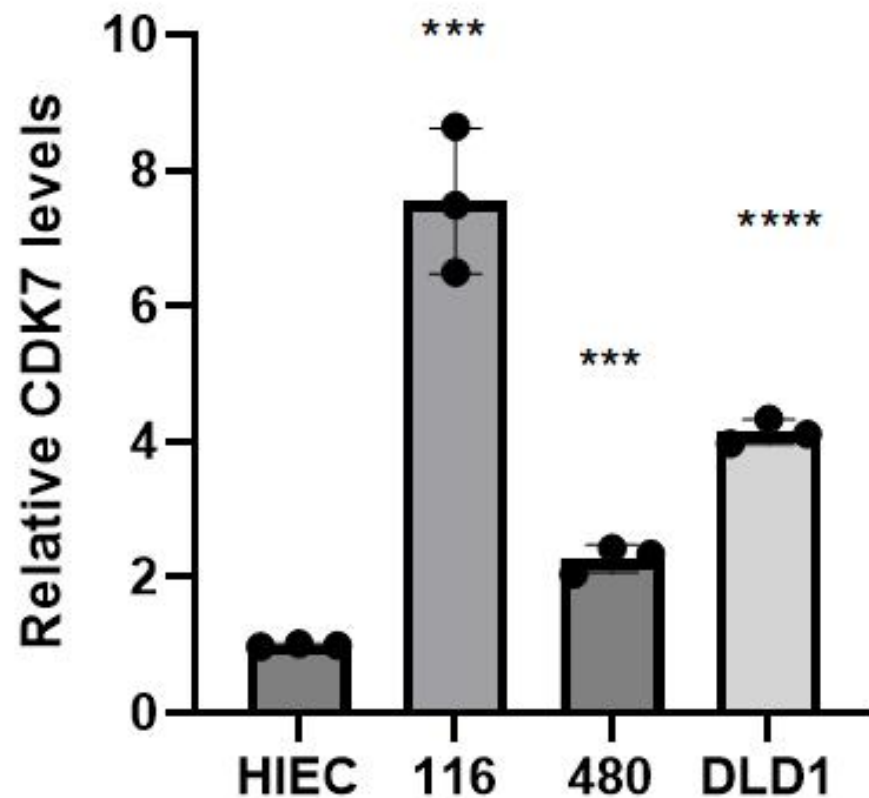


目标基因高表达与以上核心生存指标 (风险比 > 1)

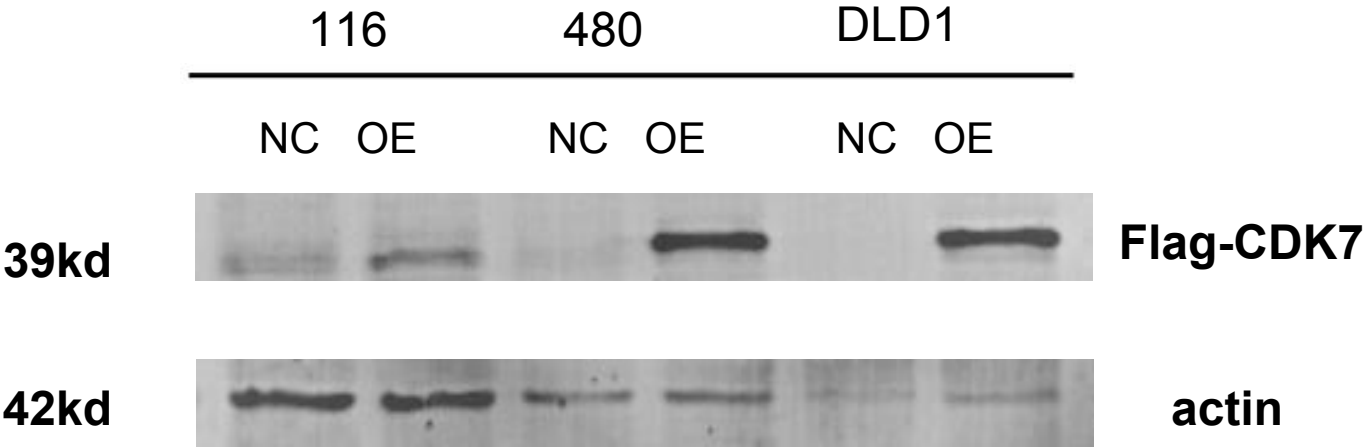
和预后相关性不大

接受 FOLFOX4 化疗的患者中，治疗无响应 (NR) 组的 CDK7 表达显著高于有响应 (R) 组

不同细胞系中CDK7的相对表达量



wb验证转染效果



EDU

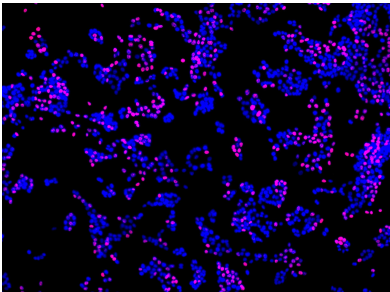
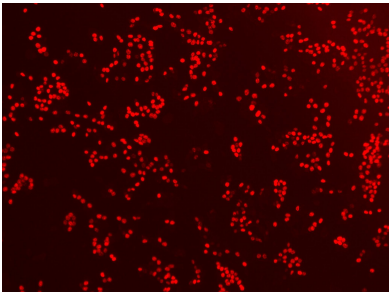
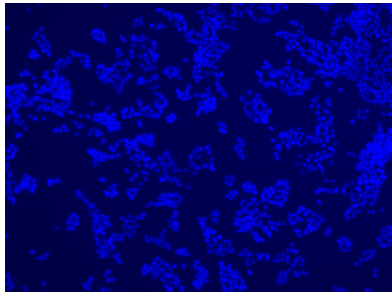
HCT116

Hoechst33342

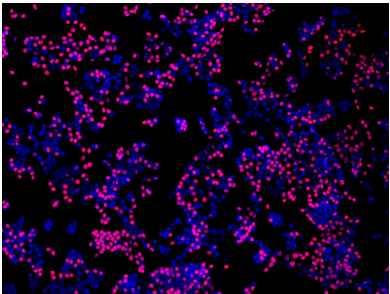
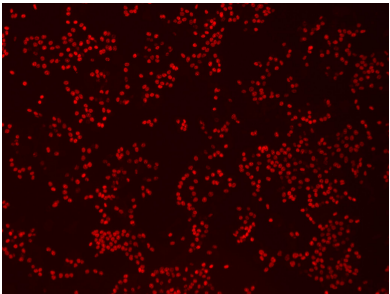
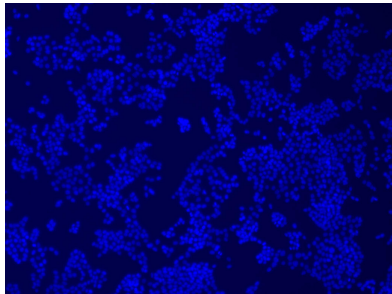
EDU

Merge

NC



OE -Flag-CDK7



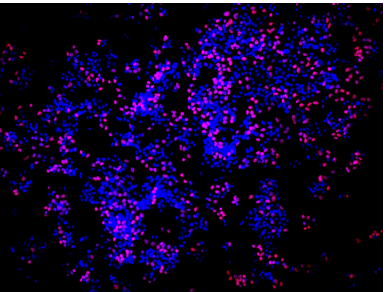
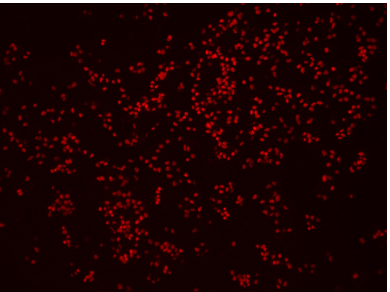
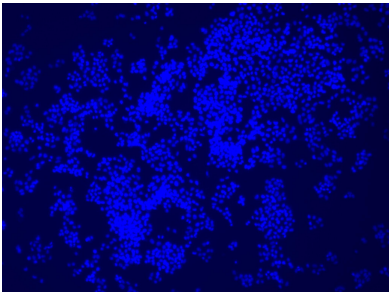
SW480

Hoechst33342

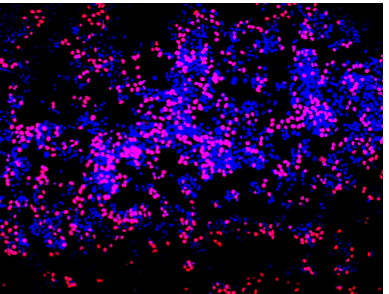
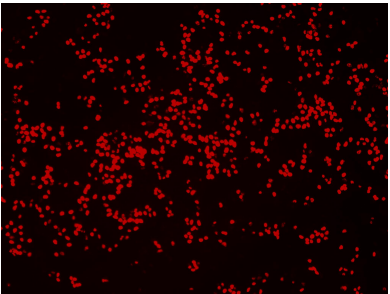
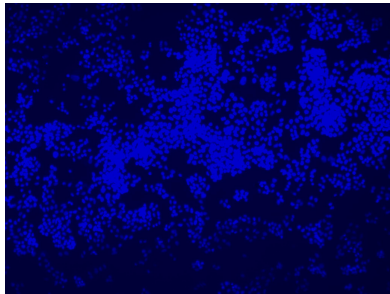
EDU

Merge

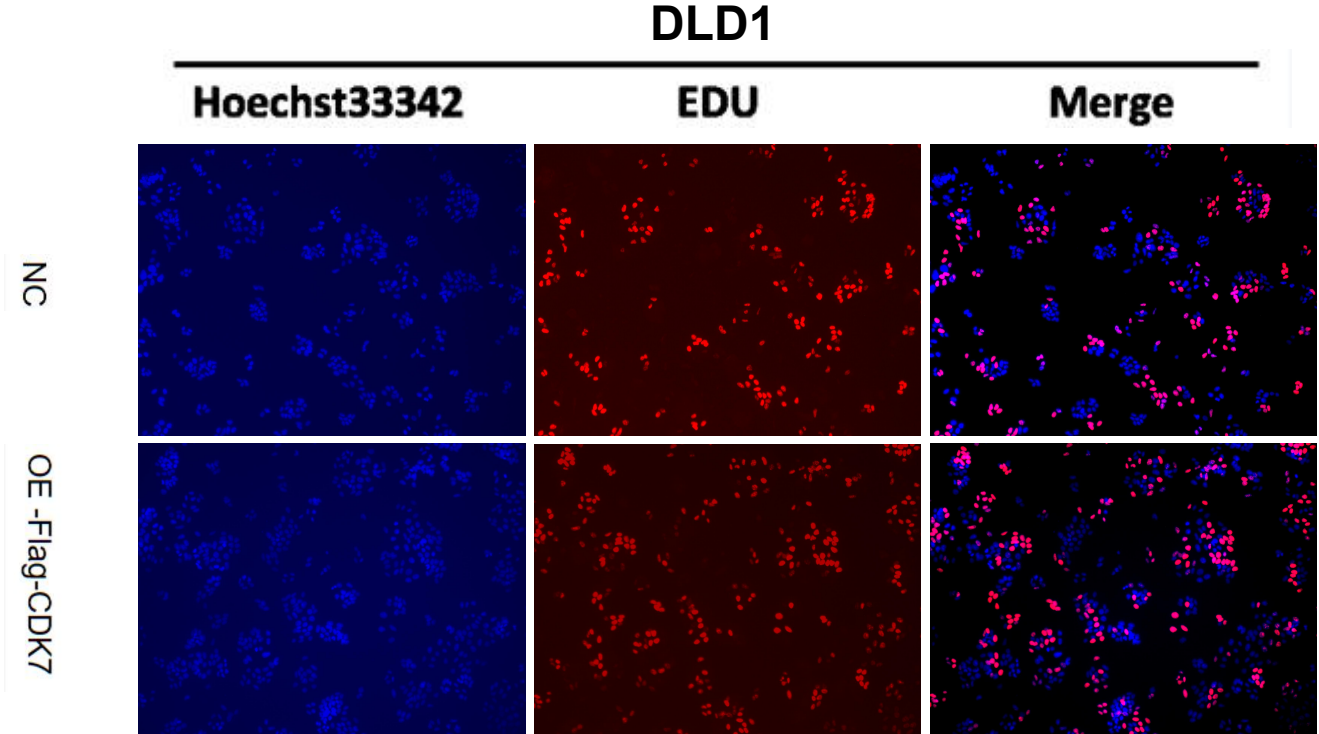
NC



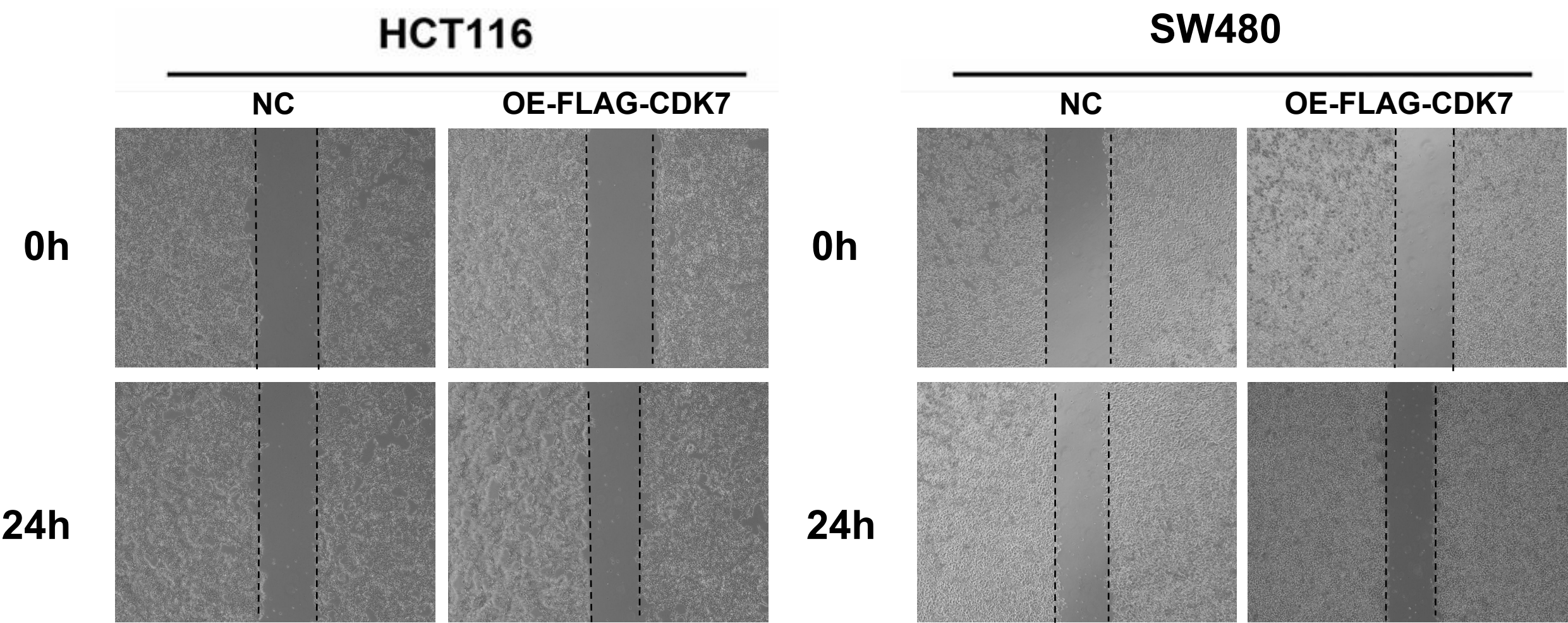
OE -Flag-CDK7



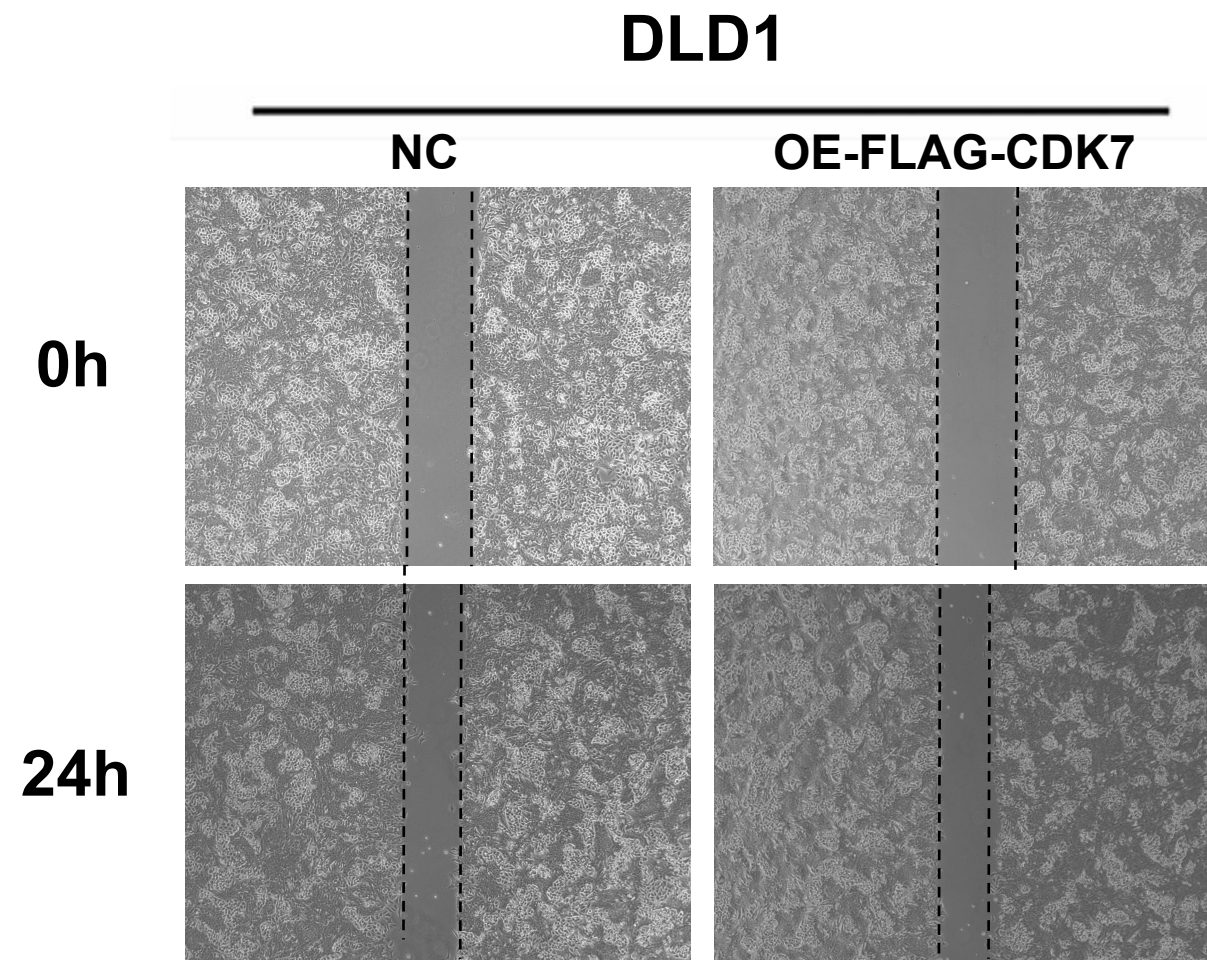
EDU



划痕

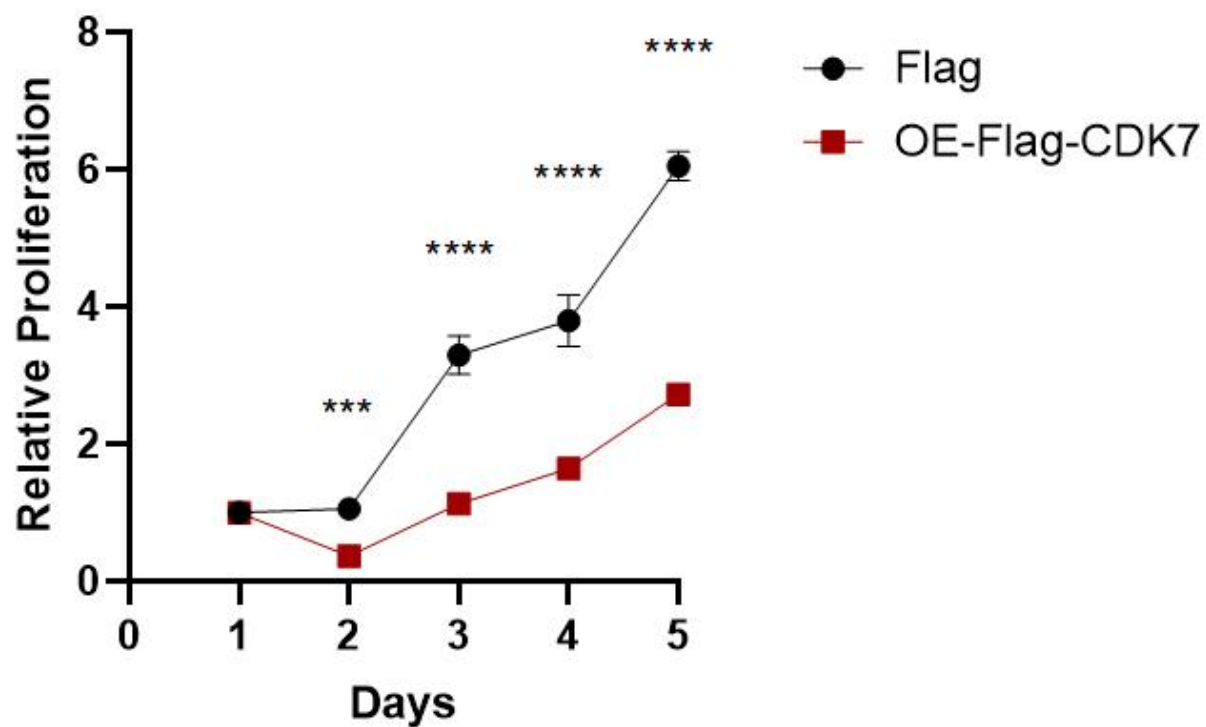


划痕

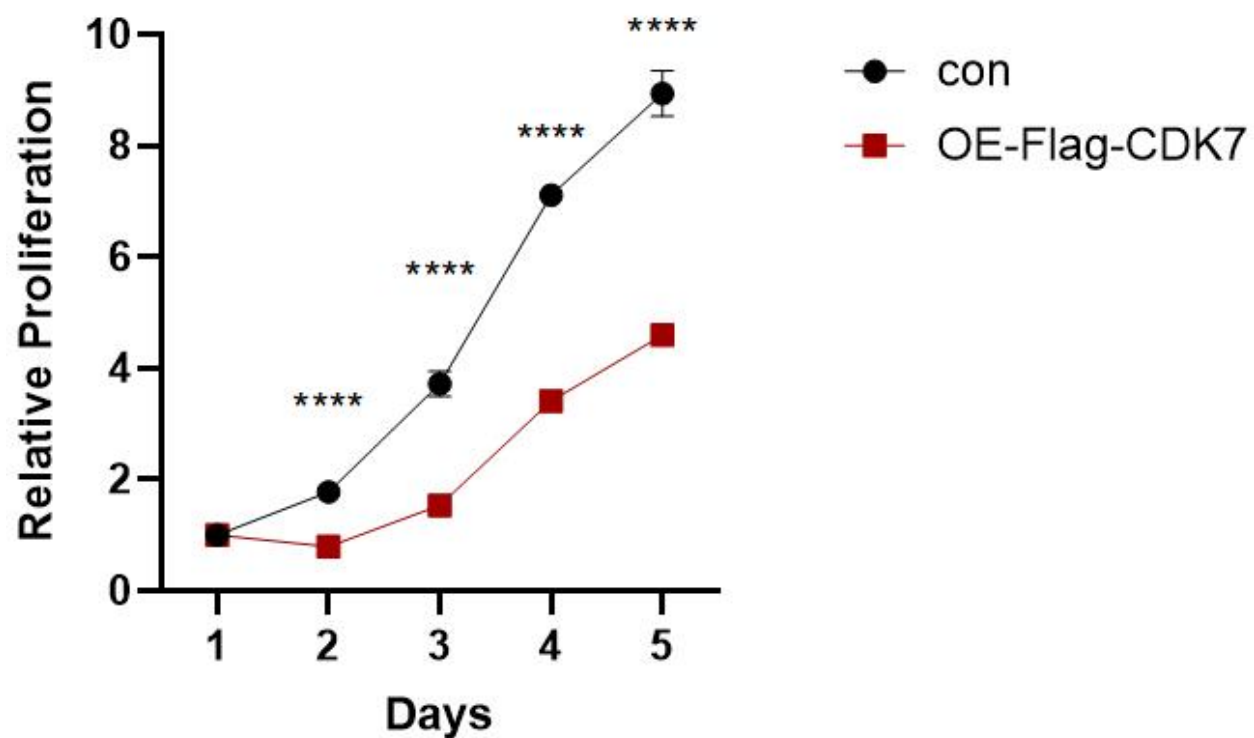


CCK8

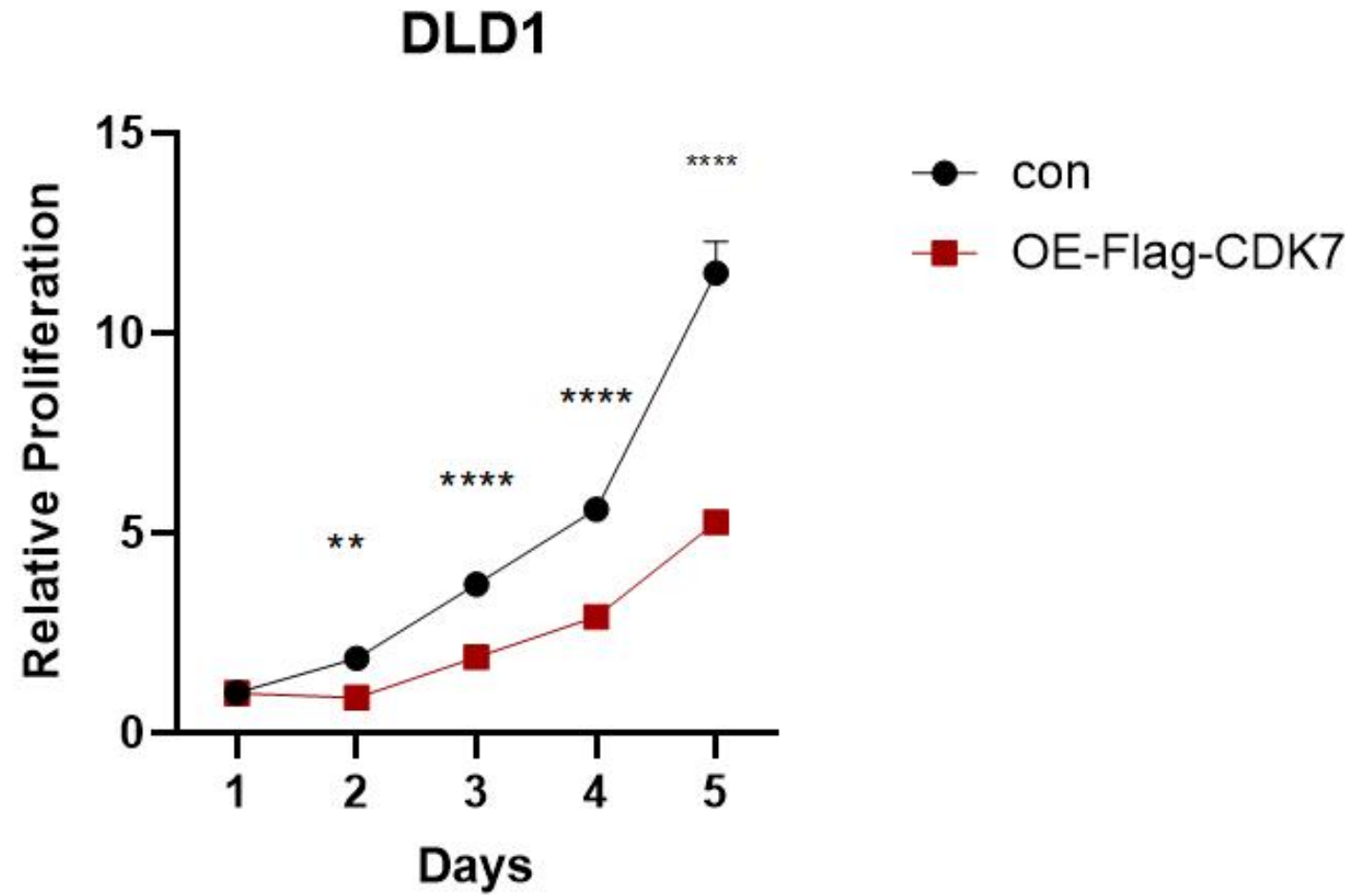
HCT116

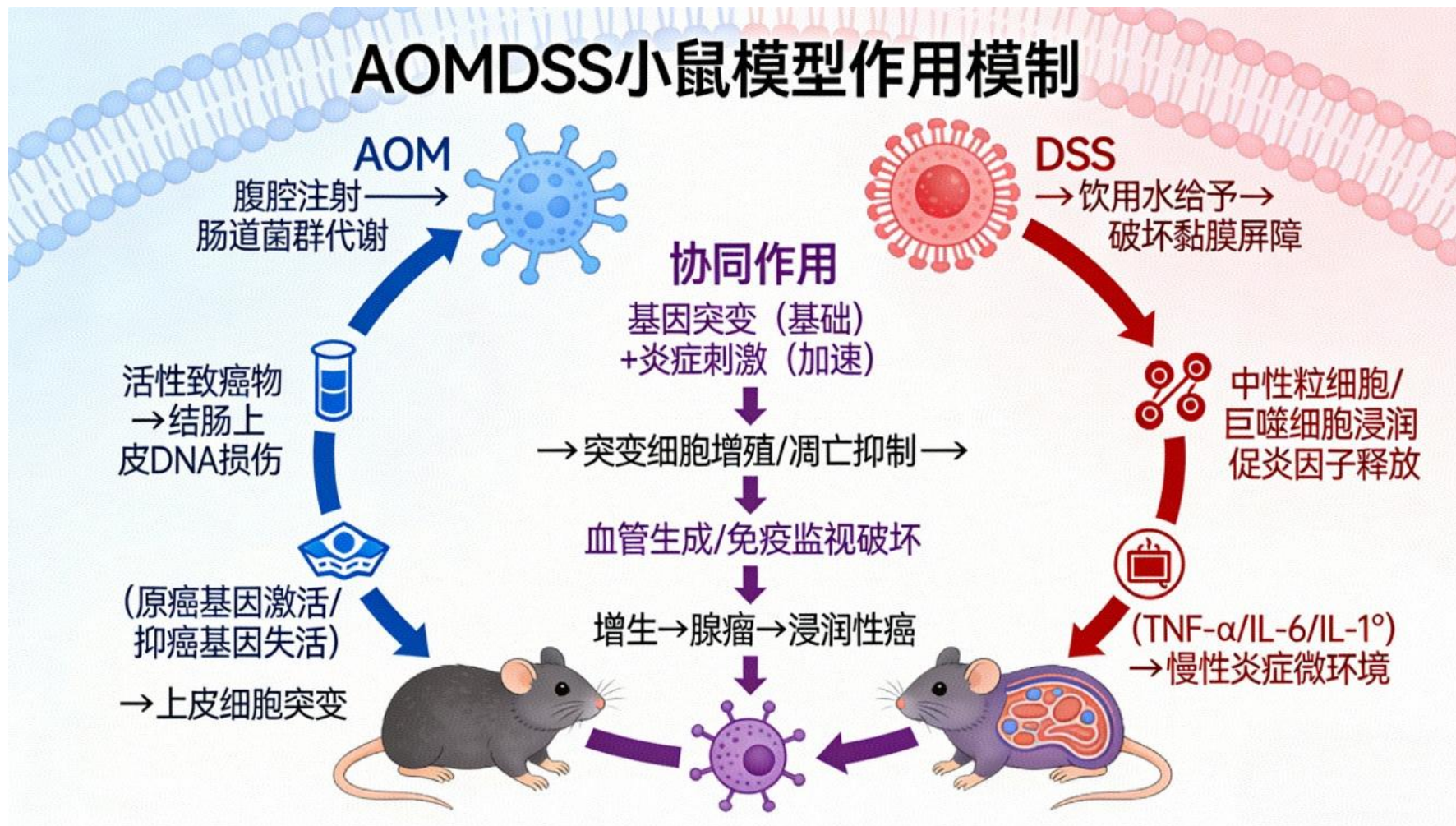


SW480



CCK8



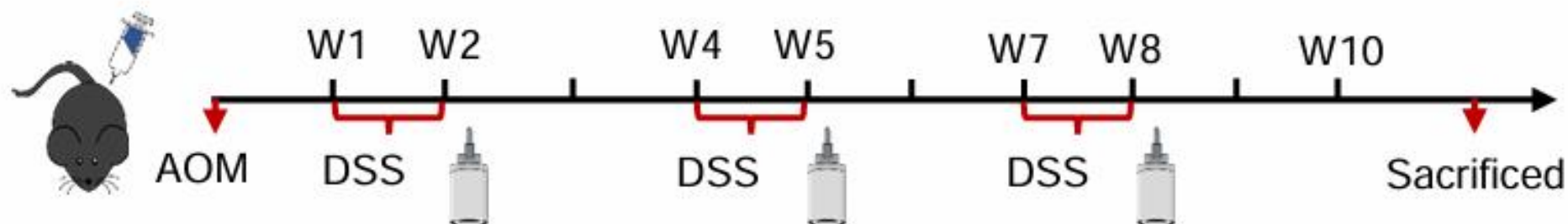


化学诱导 + 炎症驱动

原理：DSS诱导的炎症微环境会促进AOM诱导的恶变细胞的增殖、存活，抑制其凋亡，最终使结肠黏膜从炎症 $\rightarrow$ 增生 $\rightarrow$ 异型增生 $\rightarrow$ 腺瘤 $\rightarrow$ 腺癌逐步发展，完整模拟结直肠癌的炎癌转变进程

AOM-DSS肿瘤小鼠模型生成示意图

a



ROS-induced cytosolic release of mitochondrial PGAM5 promotes colorectal cancer progression by interacting with MST3. Nat Commun. 2025 Feb 6;16(1):1406. doi: 10.1038/s41467-025-56444-2IF: 15.7 Q1 . PMID: 39915446; PMCID: PMC11802746.

AOM and DSS treatment

8周龄雄性小鼠(C57BL/6)腹腔注射AOM (Sigma-Aldrich), 剂量为10 mg/kg体重。注射后1周, 小鼠饮水中加入2% DSS(分子量:36000 - 50000,MP Biomedicals) 1周, 正常饮水2周。小鼠共接受三个周期的DSS。在第三轮DSS饲喂后第7天维持正常饮水量4~6周。处死小鼠, 收集结肠远端组织, 评估肿瘤数量和面积。

原本创新点是在细胞周期蛋白和炎癌免疫之间的调控
现往线粒体自噬上靠

1. 周期流式 有效果跑周期蛋白
2. 转敲低质粒 线粒体免疫荧光看CDK7对线粒体功能的影响
3. 结晶紫
4. ROS
5. 线粒体自噬（P62 LC3）表达情况
6. 线粒体免疫荧光已在做 CCK8结果在重复中

结合已有EDU和CCK8结果

EDU↑+CCK8↓——增殖快，死的更快——细胞释放大量线粒体DNA+活性氧——激活cGAS——STING天然免疫通路

CDK7——周期加速—1.—线粒体应激—2.—线粒体自噬失衡—3.—细胞存活率下降+炎症激活——炎癌转变

体内：

1.线粒体功能检测：mtROS↑,JC-1↓,ATP↓

2.线粒体自噬标志：wb(LC3- II / I ↑P62↓自噬激活，PINK/Parkin↑

3.炎症上游：cGAS-STING-炎症-预后差

动物模型：AOM/DSS模型：CDK7过表达-线粒体损伤增多-炎症-腺瘤/癌

cGAS——STING天然免疫通路

cGAS（DNA感受器）结合细胞质里出现外源DNA / 线粒体漏出来的DNA，产生cGAMP，cGAMP激活STING，从而激活IRF3 / NF-κB，最终放出：IFN-β（I型干扰素）IL-6、TNF-α等炎症因子